

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

(Start the day with the name of Allah)

Welcome To Virtual
University



VU KMS

اسلام علیکم:

امید کرتا ہوں سب خیریت سے ہوں گئے۔ آنے والے تمام نیو طالب علم کو میں ورچوئل یونیورسٹی آف پاکستان میں خوش آمدید کرتا ہوں۔
اگر کسی سٹوڈنٹ کو کسی بھی قسم کی ہیلپ چاہے ہو تو ہم سے رابطہ کر سکتا ہے جیسا کہ: پاسٹ پیپرز، کورس سلیکشن، پروجیکٹ سلیکشن، ڈیٹ شیٹ،
فیس پے اور چینج سٹڈی پروگرام وغیرہ۔

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اگر کوئی سٹوڈنٹ کسی مجبوری کی وجہ سے یا جاب ہو لڈر ہونے کی وجہ سے اپنا ایل ایم ایس خود ہینڈل نہیں کر سکتا تو وہ ہم سے پیڈ سروس حاصل کر
سکتا ہے کوشش کرے کے آپ ہر چیز خود اٹیمپٹ کریں اگر نہیں کر سکتے تو سروس حاصل کریں۔

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انشاء اللہ آپ ہماری سروس سے مطمئن ہوں گے۔

مزید اگر کسی رہنمائی کی ضرورت ہو تو وہ رابطہ کر سکتا ہے۔
Whatsapp: 03066295623 , 03404643855

Question # 30 of 30 (Start time: 10:23:37 AM, 21 June 2021)

Total Marks: 1

if $Y=C$ (a constant) then the average rate of change of y with respect to x over the interval $[0,1]$ is

Select the correct option

☒	1	Solved by: Waqas Ahmed Whatsapp: 03066295623
☐	None of these	
☐	0	
☐	36	

Question # 1 of 30 (Start time: 10:02:56 AM, 21 June 2021)

Total Marks:

$$\log_b\left(\frac{1}{c}\right) =$$

Select the correct option

 Reload Math Equations



$$1 - \log_b c$$



$$\frac{1}{\log c}$$



Solved by: Waqas Ahmed
Whatsapp: 03066295623

$$-\log_b c$$



$$\log_b c$$

Question # 2 of 30 (Start time: 10:03:08 AM, 21 June 2021)

Total Marks: 1

What is the derivative of $2x^2 \sin(x^5)$?

Select the correct option

[Reload Math Equations](#)

None of these

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$$10x^2 \cos(x^5)$$



$$- 10x^2 \cos(x^5)$$



$$2x^2 \cos(x^5)$$

Question # 3 of 30 (Start time: 10:03:52 AM, 21 June 2021)

Total Marks: 1

The functions $f(x) = 2x + x^3$ and $f(t) = 2t + t^3$ represent the different functions.

Select the correct option

[Reload Math Equations](#)

False



Solved by: Waqas Ahmed
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True



Question # 4 of 30 (Start time: 10:04:33 AM, 21 June 2021)

Total Marks: 1

Which of the following is y-intercept of a line represented by the equation:
 $y = -4x + 17$

Select the correct option



17

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1



-4 / 17



-4

Question # 5 of 30 (Start time: 10:05:20 AM, 21 June 2021)

Total Marks: 1

Sinx, Cosx and Tanx are called _____ functions.

Select the correct option



Logarithmic



Quadratic



Exponential



Trigonometric

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Question # 6 of 30 (Start time: 10:05:47 AM, 21 June 2021)

Total Marks: 1

Which of the following is distance between the points (2, 1) and (1, 1)?

Select the correct option



(1, 0)



1

Solved by: Waqas Ahmed
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Square root of (13)



-1

Question # 7 of 30 (Start time: 10:06:33 AM, 21 June 2021)

Total Marks: 1

$$(f.g)' = f'.g - f.g'$$

Select the correct option

☒ False

Solved by: Waqas Ahmed
Whatsapp: 03066295623

☐ True

Question # 8 of 30 (Start time: 10:07:07 AM, 21 June 2021)

Total Marks: 1

Which of the followings is derivative of $f(x) = 4x^3$?

Select the correct option

[Reload Math Equations](#)

$$f'(x) = 12x$$



$$f'(x) = x^4$$



$$f'(x) = 12x^3$$



Solved by: Waqas Ahmed
Whatsapp: 03066295623

$$f'(x) = 12x^2$$

Question # 9 of 30 (Start time: 10:07:42 AM, 21 June 2021)

Total Marks: 1

Usually the number that signifies the idea of $f(x)$ being as close to limit L as want to be must be a/an

Select the correct option



Natural number



Small positive number.

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Whatsapp: 03066295623



Integer



Small negative number.

Question # 10 of 30 (Start time: 10:08:25 AM, 21 June 2021)

Total Marks: 1

Let two lines have slope a_1 and a_2 respectively. If $a_1 = a_2$ then the two lines are _____

Select the correct option



Perpendicular to each other



Parallel to x-axis



Parallel to each other

Solved by: Waqas Ahmed

Whatsapp: 03066295623



Parallel to y-axis

Question # 11 of 30 (Start time: 10:09:10 AM, 21 June 2021)

Total Marks: 1

If a function f has a relative extrema at x_0 , then

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Reload Math Equations

Select the correct option



either $f'(x_0) = 0$ or f is not differentiable at x_0



$f'(x_0) > 0$



f is differentiable at x_0



$f'(x_0) \leq 0$

Question # 12 of 30 (Start time: 10:10:24 AM, 21 June 2021)

Total Marks: 1

Derivative of a straight line parallel to y-axis is----

Select the correct option

☒	Infinity	Solved by: Waqas Ahmed Whatsapp: 03066295623
☐	+1	
☐	Zero	
☐	-1	

Question # 13 of 30 (Start time: 10:11:11 AM, 21 June 2021)

Total Marks: 1

A function $f(x) = x / |x|$ is

Select the correct option

☒ continuous at $x = 1$

Solved by: Waqas Ahmed
Whatsapp: 03066295623

☐ not continuous at $x = 1$

Question # 14 of 30 (Start time: 10:12:13 AM, 21 June 2021)

Total Marks: 1

Which of the following is y-intercept of a line represented by equation, $5y = -2x + 16$

Select the correct option



5



16 / 5

Solved by: Waqas Ahmed
Whatsapp: 03066295623



2



8

Question # 15 of 30 (Start time: 10:13:04 AM, 21 June 2021)

Total Marks: 1

The functions $\sin x$ and $\cos x$ are

Select the correct option

☒ Continuous

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☐ Discontinuous

Question # 16 of 30 (Start time: 10:13:38 AM, 21 June 2021)

Total Marks: 1

Every real number corresponds to _____ on the co-ordinate line

Select the correct option



None of these



Infinite number of points



A unique point



Two points (one positive and one negative)

Question # 17 of 30 (Start time: 10:14:08 AM, 21 June 2021)

Total Marks: 1

 $d(\tan x)/dx = \dots\dots\dots$

Select the correct option



Square of Cosx



Square of secx

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Question # 18 of 30 (Start time: 10:14:33 AM, 21 June 2021)

Total Marks: 1

$f(x) = \cos x$ is differentiable in the interval _____.

Select the correct option

 $(-\infty, 0]$  $[0, +\infty)$  $(-\infty, +\infty)$

All are true

Solved by: Waqas Ahmed

Whatsapp: 03066295623

Question # 19 of 30 (Start time: 10:15:23 AM, 21 June 2021)

Total Marks: 1

The implicit differentiation of $y^2 = x$, gives $\frac{dy}{dx} = \underline{\hspace{2cm}}$.

Select the correct option

[Reload Math Equations](#)

$\frac{1}{2x}$



1



$\frac{1}{2y}$

Solved by: Waqas Ahmed
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$2y \frac{dy}{dx}$

Question # 20 of 30 (Start time: 10:16:29 AM, 21 June 2021)

Total Marks: 1

$$\lim_{x \rightarrow +\infty} \left(\frac{9x^2}{3x^3} \right) =$$

Select the correct option

[Reload Math Equations](#) $-\infty$ 

0

Solved by: Waqas Ahmed
Whatsapp: 03066295623



3

 $+\infty$

Question # 21 of 30 (Start time: 10:17:15 AM, 21 June 2021)

Total Marks: 1

Which of the following is equation of a line whose slope is 7 and y-intercept is 3?

Select the correct option



$$y = 7x + 3$$

Solved by: Waqas Ahmed

Whatsapp: 03066295623



$$y = 3x + 7$$



$$3y = x + 7$$



$$7y = x + 3$$



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اسلام علیکم:

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اگر کسی سٹوڈنٹ کو کسی بھی قسم کی ہیلپ چاہے ہو تو ہم سے رابطہ کر سکتا ہے جیسا کہ: پاسٹ پیپرز، کورس سلیکشن، پروجیکٹ سلیکشن، ڈیٹ شیٹ،
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سکتا ہے کوشش کرے کے آپ ہر چیز خود اٹھیمپٹ کریں اگر نہیں کر سکتے تو سروس حاصل کریں۔

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مزید اگر کسی رہنمائی کی ضرورت ہو تو وہ رابطہ کر سکتا ہے۔

Question # 22 of 30 (Start time: 10:18:00 AM, 21 June 2021)

Total Marks: 1

$|x^2-9|=.....$

Select the correct option



$|x+3|^2$



$|x-3||x+3|$

Solved by: Waqas Ahmed
Whatsapp: 03066295623



$|x-3|^2$



$|x+3||x+3|$

Question # 23 of 30 (Start time: 10:18:51 AM, 21 June 2021)

Total Marks: 1

$$\lim [f(x)]^n =$$

Select the correct option

[Reload Math Equations](#)

$$[\lim f(x)]^n$$

Solved by: Waqas Ahmed
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$$[\lim f(x)]$$



$$n[\lim f(x)]$$



$$[f(x)]^n$$

Question # 24 of 30 (Start time: 10:19:31 AM, 21 June 2021)

Total Marks: 1

The equation of tangent line at point (x_0, y_0) is $y - y_0 = m \tan(x - x_0)$

Select the correct option

False

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True

Question # 25 of 30 (Start time: 10:20:15 AM, 21 June 2021)

Total Marks: 1

$$\log_b \frac{M}{N} = \underline{\hspace{2cm}}.$$

Select the correct option

[Reload Math Equations](#)

$$- \log_b N$$



Solved by: Waqas Ahmed
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$$\log_b M - \log_b N$$



$$\log_b M + \log_b N$$



$$M \log_b N$$

Question # 26 of 30 (Start time: 10:20:50 AM, 21 June 2021)

Total Marks: 1

Which of the following is slope of a line segment joining the points $(-1, 4)$ and $(3, 2)$?

Select the correct option

☒	-1 / 2	Solved by: Waqas Ahmed Whatsapp: 03066295623
☐	3	
☐	Undefined	
☐	-1 / 3	

Question # 27 of 30 (Start time: 10:21:35 AM, 21 June 2021)

Total Marks: 1

Which of the following is derivative of $f(x) = 3x$?

Select the correct option

[Reload Math Equations](#)

$$f'(x) = y$$



$$f'(x) = x$$



$$f'(x) = x^2$$



Solved by: Waqas Ahmed
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$$f'(x) = 3$$

Question # 28 of 30 (Start time: 10:22:09 AM, 21 June 2021)

Total Marks: 1

If a function is differentiable at a point then it is continuous at that point. The converse is

Select the correct option



True



False

Solved by: Waqas Ahmed
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Question # 29 of 30 (Start time: 10:23:01 AM, 21 June 2021)

Total Marks: 1

Which of the following is x - intercept of a parabola represented by the equation : $y = x^2 + 2x$

Select the correct option

[Reload Math Equations](#)

1 only



0 and -2

Solved by: Waqas Ahmed

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1 and 2



0 only

Question # 10 of 30 (Start time: 09:07:24 AM, 21 June 2021)

Total Mark

Derivative of a graph of straight line bisecting the second and fourth quadrants is-----

Select the correct option



Infinity



-1

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Zero



+1

Question # 11 of 30 (Start time: 09:08:11 AM, 21 June 2021)

Total Marks: 1

If the function f and g are continuous at c , then
 $f \circ g$ is at c .

Select the correct option



Discontinuous



Continuous

Solved by: Waqas Ahmed
Whatsapp: 03066295623

Question # 12 of 30 (Start time: 09:08:34 AM, 21 June 2021)

Total Marks: 1

The implicit differentiation of $y^2 = x$, gives $\frac{dy}{dx} = \underline{\hspace{2cm}}$.

Select the correct option

[Reload Math Equations](#)

$\frac{1}{2x}$



$2y \frac{dy}{dx}$



1



$\frac{1}{2y}$

Solved by: Waqas Ahmed
Whatsapp: 03066295623

Question # 13 of 30 (Start time: 09:09:02 AM, 21 June 2021)

Total Marks: 1

$$\log_a a =$$

Select the correct option

[Reload Math Equations](#)

$$a \log_a 1$$



$$1$$



$$0$$

Solved by: Waqas Ahmed
Whatsapp: 03066295623



$$a$$

Question # 14 of 30 (Start time: 09:09:34 AM, 21 June 2021)

Total Marks: 1

If $y=f(x)$ then instantaneous rate of change of y with respect to x at the point x_0 is given by slope of secant line.

Select the correct option

False



Solved by: Waqas Ahmed
Whatsapp: 03066295623

True



Question # 15 of 30 (Start time: 09:10:02 AM, 21 June 2021)

Total Marks: 1

Chain rule says that if 'g' is differentiable at a point x and 'f' is differentiable at the point $g(x)$ then composition of $f(g(x))$ is differentiable at the point 'x'.

Select the correct option

☒ True

Solved by: Waqas Ahmed
Whatsapp: 03066295623

☐ False

Question # 16 of 30 (Start time: 09:10:38 AM, 21 June 2021)

Total Marks: 1

$$\text{If } u(t) = 4 \cos t, \text{ then } u'(\frac{\pi}{2}) = \dots$$

Select the correct option

[Reload Math Equations](#)

2



9



4



-4

Solved by: Waqas Ahmed
Whatsapp: 03066295623

Question # 17 of 30 (Start time: 09:11:06 AM, 21 June 2021)

Total Marks: 1

Let $f(x) = 2$ and $g(x) = x^2 + 1$ then $(f - g)(x) = ?$

Select the correct option

[Reload Math Equations](#)

$2 - x^2$



$1 + x^2$



$3 - x^2$



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$1 - x^2$

Question # 18 of 30 (Start time: 09:11:47 AM, 21 June 2021)

Total Marks: 1

If a function is ONE to ONE then it has an inverse function.

Select the correct option

True



Solved by: Waqas Ahmed
Whatsapp: 03066295623

False



Question # 19 of 30 (Start time: 09:12:14 AM, 21 June 2021)

Total Marks: 1

Which of the following is midpoint of the line segment joining points $(-2, 1)$ and $(2, -2)$?

Select the correct option

 $(0, -1/2)$

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 $(-4, 3)$  $(4, -3)$  $(2, -3/2)$

Question # 21 of 30 (Start time: 09:13:13 AM, 21 June 2021)

Total Marks: 1

Suppose a and b are any two positive real numbers.
If $a < b$ then _____

Select the correct option



$$(1/a) < (1/b)$$



$$(1/a) > (1/b)$$

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Question # 22 of 30 (Start time: 09:13:48 AM, 21 June 2021)

Total Marks: 1

What is the derivative of $\sin(x)$ at $x = \pi/2$

Select the correct option

[Reload Math Equations](#)

0.707



0

Solved by: Waqas Ahmed
Whatsapp: 03066295623



None of these



1

Question # 23 of 30 (Start time: 09:14:31 AM, 21 June 2021)

Total Marks: 1

Let a function f is continuous on $[a, b]$ and differentiable on (a, b) .
If _____ for all x in (a, b) , then f is increasing on $[a, b]$.

Select the correct option

[Reload Math Equations](#)

$f'(x) < 0$



$f(x) < 0$

Solved by: Waqas Ahmed
Whatsapp: 03066295623



$f'(x) > 0$



$f'(x) = 0$

Question # 24 of 30 (Start time: 09:14:44 AM, 21 June 2021)

Total Marks: 1

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} =$$

Select the correct option

[Reload Math Equations](#)

-1

Solved by: Waqas Ahmed
Whatsapp: 03066295623



Undefined



1



0

Question # 25 of 30 (Start time: 09:14:48 AM, 21 June 2021)

Total Marks: 1

The inequality,
 $\{1 / x\} < 1$
can be simplified to which of the following?

Select the correct option



$$(1 - x) / x < 0$$



$$(1 - x) > 0$$



$$(1 - x) / x > 0$$

Solved by: Waqas Ahmed
Whatsapp: 03066295623



$$(1 - x) < 0$$

Question # 26 of 30 (Start time: 09:14:54 AM, 21 June 2021)

Total Marks: 1

The derivative of $(x+1)^{1/2}$ is

Select the correct option



$\frac{1}{3}\{(x+1)^{1/2}\}$

Solved by: Waqas Ahmed
Whatsapp: 03066295623



$x+1$



$\frac{1}{2}x$



$\frac{1}{2}\{(x+1)^{-1/2}\}$

Question # 27 of 30 (Start time: 09:15:00 AM, 21 June 2021)

Total Marks: 1

What is the value of the $\sin 3x/6x$; as x approaches to zero.?

Select the correct option



pi



1



1/2

Solved by: Waqas Ahmed

Whatsapp: 03066295623



0

Question # 28 of 30 (Start time: 09:15:40 AM, 21 June 2021)

Total Marks: 1

Which of the following is an example of set?

Select the correct option



Collection of planets in our solar system.

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Collection of temperature.



Collection of time.



Collection of good student in a class.

Question # 29 of 30 (Start time: 09:16:11 AM, 21 June 2021)

Total Marks: 1

$$\log_b\left(\frac{1}{c}\right) =$$

Select the correct option

[Reload Math Equations](#)

$$\frac{1}{\log c}$$



$$1 - \log_b c$$



Solved by: Waqas Ahmed
Whatsapp: 03066295623

$$-\log_b c$$



$$\log_b c$$

Question # 30 of 30 (Start time: 09:16:44 AM, 21 June 2021)

Total Marks: 1

If a car travels 75 miles over a straight line in a 3-hour period, then we say that the average velocity of the car is.....miles per hour

Select the correct option



75

Solved by: Waqas Ahmed
Whatsapp: 03066295623

 $3/75$  $75/3$ 

3.75

Question # 2 of 30 (Start time: 09:43:15 AM, 21 June 2021)

Total Marks: 1

The functions $\sin x$ and $\cos x$ are

Select the correct option



Discontinuous



Continuous

Solved by: Waqas Ahmed

Whatsapp: 03066295623

Question # 3 of 30 (Start time: 09:43:43 AM, 21 June 2021)

Total Marks: 1

$$(f.g)' = f'.g - f.g'$$

Select the correct option

False



Solved by: Waqas Ahmed
Whatsapp: 03066295623

True



Question # 4 of 30 (Start time: 09:44:09 AM, 21 June 2021)

Total Marks: 1

Derivative of difference equal to

Select the correct option

☒ Difference of the derivatives

Solved by: Waqas Ahmed

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☐ Sum of the derivatives

Question # 5 of 30 (Start time: 09:44:46 AM, 21 June 2021)

Total Marks: 1

If $f(x) = x^2$ and $g(x) = \tan^2 x$, then the derivative of $f(x) \times g(x)$ is

Select the correct option

[Reload Math Equations](#)

$$x^2 \tan(x) \sec^2(x) + 2x \tan^2(x).$$



None of these.



$$2x^2 \tan(x) \sec^2(x) + 2x \tan^2(x).$$



$$x^2 \tan(x) \sec(x) + 2x \tan^2(x).$$

Question # 6 of 30 (Start time: 09:45:22 AM, 21 June 2021)

Total Marks: 1

If $w = \sin x$ and $x = t$ then $\frac{dw}{dt} = \dots\dots\dots$

Select the correct option

[Reload Math Equations](#)

Solved by: Waqas Ahmed
Whatsapp: 03066295623

 $\cos t$  $\cos x$  $-\cos x$  $-\cos t$

Question # 7 of 30 (Start time: 09:45:53 AM, 21 June 2021)

Total Marks: 1

Let $f(x) = x - 3$ and $g(x) = x^3$ then $(f - g)(x) = ?$

Select the correct option

[Reload Math Equations](#)

$$x^3 + x - 3$$



Solved by: Waqas Ahmed
Whatsapp: 03066295623

$$-x^3 + x - 3$$



$$-x^3 + x + 3$$



$$-x^3 - x - 3$$

Question # 8 of 30 (Start time: 09:46:22 AM, 21 June 2021)

Total Marks:

If $y=2x$ then instantaneous rate of change of 'y' w.r.t 'x' at 'x =-2' is

Select the correct option



1



-1



2

Solved by: Waqas Ahmed

Whatsapp: 03066295623



3

Question # 9 of 30 (Start time: 09:46:50 AM, 21 June 2021)

Total Marks: 1

Which of the following is slope of a line segment joining the points $(4, 3)$ and $(-2, 3)$?

Select the correct option



5



3

 $1/7$ 

0

Solved by: Waqas Ahmed
Whatsapp: 03066295623

Question # 10 of 30 (Start time: 09:47:28 AM, 21 June 2021)

Total Marks: 1

A differentiable function is always continuous.

Select the correct option



False



True

Solved by: Waqas Ahmed
Whatsapp: 03066295623

Question # 11 of 30 (Start time: 09:47:50 AM, 21 June 2021)

Total Marks: 1

The derivative of x^3+x^2 is _____.

Select the correct option



$3x^2-2x$



$3x^2+2x$

Solved by: Waqas Ahmed
Whatsapp: 03066295623



$3x+2$



$3x-2$

Question # 12 of 30 (Start time: 09:48:35 AM, 21 June 2021)

Total Marks: 1

A graph in the xy plane represents a function if and only if no.....intersects the graph more than once.

Select the correct option



Horizontal line



Vertical line

Solved by: Waqas Ahmed

Whatsapp: 03066295623

Question # 13 of 30 (Start time: 09:49:21 AM, 21 June 2021)

Total Marks: 1

For $z = 2xy$, where $x = \sqrt{t}$ and $y = \frac{3}{\sqrt{t}}$, use the chain rule to determine which of the following is $\frac{dz}{dt}$.

Select the correct option

[Reload Math Equations](#)

6



0

Solved by: Waqas Ahmed

Whatsapp: 03066295623



3



2

Question # 14 of 30 (Start time: 09:49:56 AM, 21 June 2021)

Total Marks: 1

$$\lim \left[\frac{f(x)}{g(x)} \right] = \frac{\lim f(x)}{\lim g(x)}; \quad \lim g(x) \text{ is}$$

Select the correct option

[Reload Math Equations](#)

Not equal to zero

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Undefined



One



Equal to zero

Question # 15 of 30 (Start time: 09:50:53 AM, 21 June 2021)

Total Marks: 1

Is the graph of equation
 $4y = 3x$
 symmetric about origin?

Select the correct option

Yes



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 Whatsapp: 03066295623

No



Question # 16 of 30 (Start time: 09:51:37 AM, 21 June 2021)

Total Marks: 1

At a ____ to any graph, a tangent line does not exist because the slopes of the secant lines do not have a two sided equal limits.

Select the correct option



Mid point



Corner point

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Question # 17 of 30 (Start time: 09:52:15 AM, 21 June 2021)

Total Marks: 1

The derivative of the difference is equal to theof the derivatives.

Select the correct option



Quotient



Sum



Product



Difference

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Question # 18 of 30 (Start time: 09:52:58 AM, 21 June 2021)

Total Marks: 1

The graph of decreasing function has tangent lines with _____ slope.

Select the correct option

[Reload Math Equations](#)

Positive



Negative



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Question # 19 of 30 (Start time: 09:53:25 AM, 21 June 2021)

Total Marks: 1

Graph of $f(x)$ is the same as the graph of $cf(x)$vertically if $0 < c < 1$

Select the correct option



Stretched



Compressed

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Question # 20 of 30 (Start time: 09:54:01 AM, 21 June 2021)

Total Marks: 1

If the function is differentiable at a point then it is also _____ at that point.

Select the correct option



Not defined



Continuous

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Question # 21 of 30 (Start time: 09:54:27 AM, 21 June 2021)

Total Marks: 1

What is the derivative of $\cos(2x)$ at $x = \pi/4$?

Select the correct option

[Reload Math Equations](#)

2



-2

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None of these



0

Question # 22 of 30 (Start time: 09:55:03 AM, 21 June 2021)

Total Marks: 1

Let $f(x) = x - 3$ and $g(x) = 2x$ then which of the following is domain of the function fg ?

Select the correct option

[Reload Math Equations](#)

Set of all rational numbers



Set of all real numbers

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Set of all odd numbers



Set of all even numbers

Question # 23 of 30 (Start time: 09:55:44 AM, 21 June 2021)

Total Marks: 1

The derivative of $\sin(2x)$ is

Select the correct option

 $\cos 2x$  $2\cos(x)$  $\cos x$  $\cos(2x) \cdot 2$

Solved by: Waqas Ahmed

Whatsapp: 03066295623

Question # 24 of 30 (Start time: 09:56:22 AM, 21 June 2021)

Total Marks: 1

A functionbe differentiable at the point of discontinuity.

Select the correct option



can



cannot

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Question # 25 of 30 (Start time: 09:56:48 AM, 21 June 2021)

Total Marks: 1

What is the value of the $\sin 3x/6x$; as x approaches to zero.?

Select the correct option



1

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 $1/2$ 

0

 π

Question # 26 of 30 (Start time: 09:56:53 AM, 21 June 2021)

Total Marks: 1

Square root of 2 is an example of _____

Select the correct option



Natural numbers



Irrational numbers



Rational numbers

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Integers

Question # 27 of 30 (Start time: 09:56:58 AM, 21 June 2021)

Total Marks: 1

Which of the following is distance between the points (2, 5) and (-1, 1)?

Select the correct option

☒	Square root of (37)	Solved by: Waqas Ahmed Whatsapp: 03066295623
☐	25	
☐	5	
☐	Square root of (-7)	

Question # 28 of 30 (Start time: 09:57:02 AM, 21 June 2021)

Total Marks: 1

If $f(x) = (\sec x + x^4)^5$, then the derivative of $f(x)$ is _____.

Select the correct option

[Reload Math Equations](#)

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Whatsapp: 03066295623

$$5(\sec x + x^4)^4(\sec(x) \tan(x) + 4x^3).$$



$$5(\sec x + x^4)^4.$$



None of these.



$$5(\sec x + x^4)^4(\sec(x) \tan(x) - 4x^3).$$

Question # 29 of 30 (Start time: 09:57:36 AM, 21 June 2021)

Total Marks: 1

What is the derivative of $5\sin(x) - 2\cos(x)$?

Select the correct option

 $5\cos(x) + 2\sin(x)$

Solved by: Waqas Ahmed
Whatsapp: 03066295623

 $5\cos(x) - 2\sin(x)$ 

None of these

 $5\sin(x) + 2\cos(x)$

Question # 30 of 30 (Start time: 09:58:25 AM, 21 June 2021)

Total Marks: 1

If $f(x) = x^2 \sin(2x)$, then the derivative of $f(x)$ is

Select the correct option

[Reload Math Equations](#)

$$2x^2 \cos(x) + 2x \sin(2x)$$



$$2x \cos(2x) + 2x \sin(2x).$$



None of these.



Solved by: Waqas Ahmed
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$$2x^2 \cos(2x) + 2x \sin(2x)$$

Question # 1 of 30 (Start time: 09:42:32 AM, 21 June 2021)

Total Marks: 1

The solution interval of the inequality $12 < x + 5$ is

Select the correct option

Reload Math Equations

- | | |
|----------------------------------|--|
| ☐ | $[5, 12]$ |
| ☒ | <p>Solved by: Waqas Ahmed</p> <p>Whatsapp: 03066295623</p> <p>$(7, \infty)$</p> |
| ☐ | $(-\infty, \infty)$ |
| ☐ | $(-\infty, 7)$ |